

Add-on 1 Reproduce BioLangFusion

Phase A. Its encoders (NT v2 100M, RNA-FM, ESM-2 8M), codon grid, three heads, TextCNN; published splits and recipe. Five benchmarks: Fungal, E. coli, mRNA stability, Ab1, CoV-Vac. Gate: match the published scores or record each deviation. Slots into Tier 2, Gantt 17.



Add-on 2 Change encoders only

Phase B. Same splits, grid, heads, head and evaluation; one encoder swapped at a time (CaLM, mRNA-FM, ProtBERT, other DNA encoders), then triples. Report: published, reproduced, best single, each head, each swap, gain. Stage 1 says which swaps should gain. Slots into Tier 3, Gantt 24 (now planned).



Add-on 3 LucaOne, jointly pretrained

One model for nucleic acid and protein, frozen; same folds and head on stability and GTEx. Verify input limits (1,280 tokens) and sequence types first. Compare against the best single encoder and the best fusion; report cost and truncation. Slots into Tier 2, beside Gantt 16 and 17.



Add-on 4 Full-transcript tasks

Half-life (Saluki human, mouse; LDKWK) and translation efficiency (TE human, mouse; MRL Sugimoto). Gates: UTRs and CDS coordinates present, labels distinct, grouped non-leaking splits. Compare CDS-only, CDS + 3' UTR, full transcript, + protein; gain over the best full-transcript baseline. Tier 4, Gantt 38.

Add-on 5, parallel mRFP control

Cached embeddings and labels in hand. Protein is identical by construction, so protein encoders are an invariant negative control; nucleotide encoders against GC3, codon-use and structure proxies. Tests representation specificity, not fusion. Tier 1, Gantt 18.

Add-on 6, later Structure features

Only on a task where structure plausibly acts. Protein: sequence vs + ESMFold vs + AlphaFold2 features. RNA: sequence vs + predicted pairing (RNAfold thermodynamic vs EternaFold, MXfold2). Report gains over sequence-only and over simple structure features. Conditional, Gantt 38.

Add-on 7, exploratory Multi-teacher distillation

Distill the DNA, RNA and protein teachers into one compact student per modality with a relational loss (match each teacher's cosine-similarity matrix through its own projection), then fuse the students. A new hypothesis; not implemented before the core work. Stage 1 bears on it: much teacher similarity is composition.

Add-ons to the Stage 3 and 4 design and to the Gantt chart (as of October 3; dates subject to change). Coupled Mamba stays the priority architecture (Gantt 20 to 21).

Every result reports Δ_{fusion} = Perf(fused) minus the best single encoder, confidence intervals, and the baselines B0 to B5.